Econometrics III
Bachelor in Econometrics and Data Science
Bachelor in Econometrics and Operations Research

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Vector Error Correction Models

This Course: Econometrics 3

1. Multiple equations in regression models (week 1)

2. Panel data models (weeks 2 and 3)
   - ▷ Introduction and Random Effects model
   - ▷ Fixed Effects model and some extensions

3. Vector autoregressive models (weeks 3 and 4)
   - ▷ Stable VAR(1), dynamic properties, forecasting
   - ▷ Granger causality, impulse response, estimation, Stable VAR(p)

4. Non-stationary, cointegration, vector error correction (week 5)

5. Principal components, dynamic factor models (week 6)

Non-stationary, Cointegration and Vector Error Correction Models

- VAR model and non-stationary processes
- Cointegration
- Vector Error Correction Model (VECM)
- Johansen Methodology
- Discussion

Non-stationary, Cointegration and VECM

⇒ Analyzing multiple non-stationary time series simultaneously provides powerful tools to explore *long-term dynamic inter-relations* between economic variables but is rather challenging, with the danger of "spurious regressions".

⇒ It was mainly the research on non-stationary time series by Clive Granger, David Hendry, Rob Engle and Soren Johansen that have led to the rapid development of the theory and methodology on cointegration. Granger and Engle won the Nobel Prize in Economics for their work on this topic.

⇒ VECM has become the main model for cointegration.

Literature

- Key readings:
  - H. Luetkepohl, New Introduction to Multiple Time Series Analysis
  - B. H. Baltagi, Econometric Analysis of Panel Data
- Also recommended:
  - J. D. Hamilton, Time Series Analyis
  - J. M. Wooldridge, Econometric Analysis of Cross Section and Panel Data
  - H. Pesaran, Time Series and Panel Data Econometrics

VAR(1)

The VAR(1) model with $N = 2$ is given by

$$y_t = \omega + \Phi\, y_{t-1} + \varepsilon_t, \qquad t \in \mathbb{Z},$$

and more explicitly given by

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix} + \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix},$$

where $\varepsilon_t \sim \mathrm{NID}(0, \Sigma)$ with

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix}.$$

VAR(1) model

The VAR(1) model

$$y_t = \omega + \Phi\, y_{t-1} + \varepsilon_t, \qquad t \in \mathbb{Z},$$

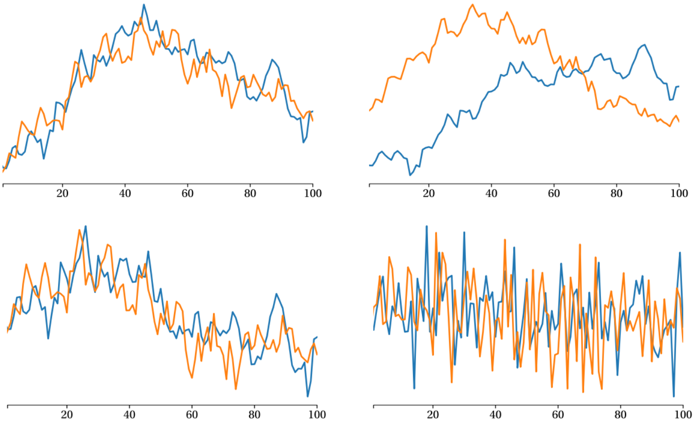
is able to represent many different time series. Let's consider the following four Φ matrices in the VAR(1) model:

$$\Phi_{[I]} = \begin{bmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{bmatrix}, \qquad \Phi_{[II]} = \begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix},$$

$$\Phi_{[III]} = \begin{bmatrix} 0.7 & 0.2 \\ 0.2 & 0.7 \end{bmatrix}, \qquad \Phi_{[IV]} = \begin{bmatrix} -0.3 & 0.3 \\ 0.1 & -0.2 \end{bmatrix}.$$

We can generate time series from these VAR(1) processes by randomly drawing ε_t's from the bivariate normal $N(0, \Sigma)$.

Simulated VAR(1) time series with $N = 2$



VAR(1) model

Let's consider these four Φ matrices in the VAR(1) model again, but now also with their eigenvalues !!

$$\Phi_{[I]} = \begin{bmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{bmatrix}, \qquad \Phi_{[II]} = \begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix},$$

$$\lambda_{[I]} = (1.0\,,\, 0.6), \qquad \lambda_{[II]} = (1.0\,,\, 1.0),$$

$$\Phi_{[III]} = \begin{bmatrix} 0.7 & 0.2 \\ 0.2 & 0.7 \end{bmatrix}, \qquad \Phi_{[IV]} = \begin{bmatrix} -0.3 & 0.3 \\ 0.1 & -0.2 \end{bmatrix},$$

$$\lambda_{[III]} = (0.9\,,\, 0.5), \qquad \lambda_{[IV]} = (-0.43\,,\, -0.06).$$

VAR(1) model

These four sets of eigenvalues

$$\lambda_{[I]} = (1.0\,,\,0.6), \qquad \lambda_{[II]} = (1.0\,,\,1.0),$$
$$\lambda_{[III]} = (0.9\,,\,0.5), \qquad \lambda_{[IV]} = (-0.43\,,\,-0.06).$$

imply that the transformed series $z_t = C_{[i]}^{-1}\,y_t$, where $C_{[i]}$ contain the eigenvectors of $\Phi_{[i]}$ for $i = I, II, III, IV$, are

Φ	$z_{1,t}$	$z_{2,t}$
$[I]$	non-stationary	stationary
$[II]$	non-stationary	non-stationary
$[III]$	persistent / stationary	stationary
$[IV]$	anti / stationary	anti / stationary

VAR(1) model

So, for z_t we have

Φ	$z_{1,t}$	$z_{2,t}$
[I]	non-stationary	stationary
[II]	non-stationary	non-stationary
[III]	persistent / stationary	stationary
[IV]	anti / stationary	anti / stationary

but since $y_t = C\, z_t$, for y_t we have

Φ	$y_{1,t}$	$y_{2,t}$
[I]	non-stationary	non-stationary
[II]	non-stationary	non-stationary
[III]	persistent / stationary	persistent / stationary
[IV]	anti / stationary	anti / stationary

VAR(1) model

Hence, these two sets of eigenvalues

$$\lambda_{[I]} = (1.0\,,\, 0.6), \qquad \lambda_{[II]} = (1.0\,,\, 1.0),$$

imply that both time series in y_t for the VAR models with corresponding Φ matrices

$$\Phi_{[I]} = \begin{bmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{bmatrix}, \qquad \Phi_{[II]} = \begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix},$$

generate non-stationary time series.

However, there is a BIG difference:

- $\Phi_{[I]}$ imply series in y_t share a <u>single</u> non-stationary "trend";
- $\Phi_{[II]}$ imply series in y_t have separate non-stationary "trends".

Non-stationary case $\Phi_{[II]} = I$

The non-stationarity of the case $\Phi_{[II]} = I$, the VAR(1) model becomes the bivariate random walk process

$$y_t = I\, y_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim \mathrm{NID}(0, \Sigma),$$

which can be treated simply by considering the first differences

$$\Delta y_t = y_t - y_{t-1} = \varepsilon_t,$$

which is a bivariate stationary process.

Analyses can be based on this model in first differences.
However, we don't learn very much interesting.
In economics we are interested in (long-run) equilibrium relationships which cannot be inferred from this setting.

Non-stationary case $\Phi_{[I]} = < 0.8, 0.2; 0.2, 0.8 >$

The non-stationary case of $\Phi_{[I]}$ is the more interesting one, both from economics and econometrics perspectives.

Economics: the two time series in y_t share the same "common stochastic trend" $z_{1,t} = z_{1,t-1} + \varepsilon^*_{1,t}$ in the decomposition $y_t = C\, z_t$, that is

$$y_{1,t} = C_{11}\, z_{1,t} + \text{stationary processes},$$
$$y_{2,t} = C_{21}\, z_{1,t} + \text{stationary processes},$$

Given that the two time series share the same trend $z_{1,t}$, we can establish a long-run equilibrium relationship between $y_{1,t}$ and $y_{2,t}$:

$$y_{1,t} - (C_{11}/C_{21})y_{2,t} = \text{stationary processes}.$$

Non-stationary case $\Phi_{[l]} =< 0.8, 0.2; 0.2, 0.8 >$

This stable long-run equilibrium relationship between $y_{1,t}$ and $y_{2,t}$:

$$y_{1,t} - \beta_c\, y_{2,t} = \text{stationary processes}, \qquad \beta_c = C_{11}/C_{21},$$

implies cointegration and has many examples in econ and finance:

- between capital and output,
- real wages and labor productivity,
- nominal exchange rates and relative prices,
- consumption and disposable income,
- long and short-term interest rates,
- money velocity and interest rates,
- price of shares and dividends,
- production and sales,
- etc. etc.

Quick return to Non-stationary case $\Phi_{[II]} = I$

In case $\lambda_1 = 1$ and $\lambda_2 = 1$, the decomposition $y_t = C\, z_t$ remains valid, that is

$$\begin{aligned} y_{1,t} =& C_{11}\, z_{1,t} + C_{12}\, z_{2,t} + \text{stationary processes}, \\ y_{2,t} =& C_{21}\, z_{1,t} + C_{22}\, z_{2,t} + \text{stationary processes}, \end{aligned}$$

The two time series in y_t now share two trends, $z_{1,t}$ and $z_{2,t}$: can we establish a long-run equilibrium relation between $y_{1,t}$ and $y_{2,t}$? Not really,

$$y_{1,t} - (C_{11}/C_{21}) y_{2,t} = (C_{12} - C_{11}\, C_{22}/C_{21})\, z_{2,t} + \text{stationary processes},$$

as, for $N = 2$, we can remove $z_{1,t}$ but NOT both $z_{1,t}$ and $z_{2,t}$.

Hence, for $N = 2$, we strictly require $\lambda_1 = 1$ and $-1 < \lambda_2 < 1$ to establish a long-run equilibrium relationship between $y_{1,t}$ and $y_{2,t}$.

Non-stationary case $\Phi_{[I]} =< 0.8, 0.2; 0.2, 0.8 >$

Econometrics: Cointegration refers to the analyis of non-stationary time series when long-run equilibrium relationships can be established. For our bivariate VAR(1) with $\Phi_{[I]}$, it is

$$y_{1,t} - \beta_c y_{2,t} = \text{stationary processes}, \qquad \beta_c = C_{11}/C_{21},$$

which only exists when one eigenvalue of Φ equals unity and the other is "stable". In summary,

- $-1 < \lambda_1 < 1$ AND $-1 < \lambda_2 < 1$:
  stationary VAR(1) process, see previous slides;
- $\lambda_1 = 1$ AND $-1 < \lambda_2 < 1$, or vice-versa :
  non-stationary VAR(1) process with cointegration;
- $\lambda_1 = 1$ AND $\lambda_2 = 1$:
  non-stationary VAR(1) becomes bivariate random walk process.

Cointegration and VECM

Cointegrated time series:

- exhibit temporary deviations from a long-run trend but
- are ultimately mean reverting to this trend which
- represent the long-run equilibrium relationship.

The Vector Error Correction Model (VECM):

- explicitly includes the deviation from the long-run relationship
- when modeling the short-term dynamics of the time series
- to push the time series towards their long-run relationship.

Error correction model

- Hence, the error correction model (ECM) is closely related to cointegration; see Davidson, Hendry, Srba, & Yeo (DHSY, 1978) and Engle & Granger (1987).
- In ECMs, the changes in one variable depend on the deviations from some equilibrium relation.
- For example, in economics consumption $y_{1,t}$ and disposable income $y_{2,t}$ are related to the equilibrium relationship

$$y_{1t} = \beta_c y_{2t}.$$

- Changes in y_{1t} and y_{2t} then depend on deviations from equilibrium in previous period, $t-1$.

Error correction models

- The basic ECM for y_{1t} and y_{2t} is then given by

$$\Delta y_{1t} = \alpha_1(y_{1,t-1} - \beta_c y_{2,t-1}) + \varepsilon_{1t},$$
$$\Delta y_{2t} = \alpha_2(y_{1,t-1} - \beta_c y_{2,t-1}) + \varepsilon_{2t}.$$

- DHSY argued that in case (y_{1t}, y_{2t}) are non-stationary, and $(\Delta y_{1t}, \Delta y_{2t})$ and $(\varepsilon_{1t}, \varepsilon_{2t})$ are stationary, then the equilibrium relationship $y_{1,t-1} - \beta_c y_{2,t-1}$ must also be stationary, provided that $(\alpha_1, \alpha_2) \neq 0$.
- It follows that ECM implicitly defines cointegration.
- Hence, ECM and cointegration are closely aligned.

Vector error correction model

The ECM

$$\Delta y_{1t} = \alpha_1(y_{1,t-1} - \beta_c y_{2,t-1}) + \varepsilon_{1t},$$
$$\Delta y_{2t} = \alpha_2(y_{1,t-1} - \beta_c y_{2,t-1}) + \varepsilon_{2t}.$$

can be expressed in vectors and matrices, we obtain the vector error correction model (VECM) as given by

$$\Delta y_t = \underbrace{\begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}}_{\alpha} \underbrace{\begin{pmatrix} 1 & -\beta_c \end{pmatrix}}_{\beta'} \underbrace{\begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix}}_{y_{t-1}} + \varepsilon_t,$$

where $y_t = (y_{1t}, y_{2t})'$ and $\varepsilon_t = (\varepsilon_{1t}, \varepsilon_{2t})'$.

This VECM has a stationary dependent variable vector Δy_t and a stationary regression variable $\beta' y_{t-1}$.

Vector error correction model

The VECM

$$\Delta y_t = \underbrace{\begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}}_{\alpha} \underbrace{\begin{pmatrix} 1 & -\beta_c \end{pmatrix}}_{\beta'} \underbrace{\begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix}}_{\mathbf{y}_{t-1}} + \varepsilon_t,$$

is written in matrix form as

$$\Delta y_t = \Pi y_{t-1} + \varepsilon_t,$$

where $\Pi = \alpha\,\beta'$ with *speed of adjustment* vector $\alpha = (\alpha_1, \alpha_2)'$ and cointegrating vector $\beta = (1, -\beta_c)'$.

$\text{Rank}(\Pi) = 1 < N$ with $N = 2$.
A necessary condition for cointegration:

<u>matrix Π is non-zero and rank-deficient</u>

Cointegrating VAR to VECM: illustration

Our simple cointegrating VAR(1) is

$$\begin{pmatrix} y_{1,t} \\ y_{2,t} \end{pmatrix} = \begin{bmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{bmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix},$$

and is turned into a VECM by

$$\begin{aligned}
\begin{pmatrix} y_{1,t} - y_{1,t-1} \\ y_{2,t} - y_{2,t-1} \end{pmatrix} &= \left\{ \begin{bmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \varepsilon_t, \\
\begin{pmatrix} \Delta y_{1,t} \\ \Delta y_{2,t} \end{pmatrix} &= \begin{bmatrix} -0.2 & 0.2 \\ 0.2 & -0.2 \end{bmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \varepsilon_t \\
&= \left[\begin{pmatrix} -0.2 \\ 0.2 \end{pmatrix} (1\,,\,-1) \right] \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \varepsilon_t, \\
\Delta y_t &= \Pi y_{t-1} + \varepsilon_t, \qquad \Pi = \alpha\,\beta'.
\end{aligned}$$

with $\alpha = (-0.2\,,\,0.2)'$ and cointegrating vector $\beta = (1\,,\,-1)'$.

Error correction models

- The basic ECM for y_{1t} and y_{2t} is given by

$$\Delta y_{1t} = \alpha_1(y_{1,t-1} - \beta_c y_{2,t-1}) + \varepsilon_{1t},$$
$$\Delta y_{2t} = \alpha_2(y_{1,t-1} - \beta_c y_{2,t-1}) + \varepsilon_{2t}.$$

- DHSY also allows changes in y_{1t} and y_{2t} to depend on past changes in both variables, for example

$$\Delta y_{1t} = \alpha_1(y_{1,t-1} - \beta_c y_{2,t-1}) + \gamma_{11,1}\Delta y_{1,t-1} + \gamma_{12,1}\Delta y_{2,t-1} + \varepsilon_{1t},$$
$$\Delta y_{2t} = \alpha_2(y_{1,t-1} - \beta_c y_{2,t-1}) + \gamma_{21,1}\Delta y_{1,t-1} + \gamma_{22,1}\Delta y_{2,t-1} + \varepsilon_{2t}.$$

- In case (y_{1t}, y_{2t}) are non-stationary, and $(\Delta y_{1t}, \Delta y_{2t})$ and $(\varepsilon_{1t}, \varepsilon_{2t})$ are stationary, then the equilibrium relationship $y_{1,t-1} - \beta_c y_{2,t-1}$ must also be stationary, provided that $(\alpha_1, \alpha_2) \neq 0$.
- Hence, also this extended ECM implicitly defines cointegration.

Vector error correction model

The ECM

$$\Delta y_{1t} = \alpha_1(y_{1,t-1} - \beta_c y_{2,t-1}) + \gamma_{11,1}\Delta y_{1,t-1} + \gamma_{12,1}\Delta y_{2,t-1} + \varepsilon_{1t},$$
$$\Delta y_{2t} = \alpha_2(y_{1,t-1} - \beta_c y_{2,t-1}) + \gamma_{21,1}\Delta y_{1,t-1} + \gamma_{22,1}\Delta y_{2,t-1} + \varepsilon_{2t}.$$

can be expressed in vectors and matrices, we obtain the vector error correction model (VECM) as given by

$$\Delta y_t = \underbrace{\begin{pmatrix}\alpha_1 \\ \alpha_2\end{pmatrix}}_{\alpha} \underbrace{\begin{pmatrix}1 & -\beta_c\end{pmatrix}}_{\beta'} \underbrace{\begin{pmatrix}y_{1,t-1} \\ y_{2,t-1}\end{pmatrix}}_{y_{t-1}} + \underbrace{\begin{pmatrix}\gamma_{11,1} & \gamma_{12,1} \\ \gamma_{21,1} & \gamma_{22,1}\end{pmatrix}}_{\Gamma} \Delta y_{t-1} + \varepsilon_t,$$

that is

$$\Delta y_t = \Pi y_{t-1} + \Gamma \Delta y_{t-1} + \varepsilon_t, \qquad \Pi = \alpha\,\beta',$$

with $\text{rank}(\Pi) = 1 < N$.

From general VAR to VECM

Let us assume a cointegrating VAR(p) model

$$y_t = \Phi_1 y_{t-1} + \ldots + \Phi_p y_{t-p} + \varepsilon_t,$$

which we verify by expressing the VAR(p) in its companion VAR(1) form and by computing the eigenvalues of its $pN \times pN$ matrix Ψ. Assume we have found $0 < m < N$ eigenvalues equal to unity, hence all time series in y_t are non-stationary.

We can express this cointegrating VAR as the general VECM

$$\Delta y_t = \Pi y_{t-1} + \Gamma_1 \Delta y_{t-1} + \ldots + \Gamma_{p-1} \Delta y_{t-p+1} + \varepsilon_t,$$

where Π and Γ_i, for $i = 1, \ldots, p-1$, are functions of $\Phi_1, \ldots, \Phi_p$ and $\text{rank}(\Pi) = r = N - m$.
This general result is obtained by recursive substitution.

Illustration: VAR(3) to VECM

We have a cointegrating VAR(3) model

$$
\begin{aligned}
y_t &= \Phi_1 y_{t-1} + \Phi_2 y_{t-2} + \Phi_3 y_{t-3} + \varepsilon_t \\
&\qquad \Rightarrow \; +\Phi_3 y_{t-2} - \Phi_3 y_{t-2} \\
&= \Phi_1 y_{t-1} + (\Phi_2 + \Phi_3) y_{t-2} - \Phi_3 \Delta y_{t-2} + \varepsilon_t \\
&\qquad \Rightarrow \; +/-(\Phi_2 + \Phi_3) y_{t-1} \\
&= (\Phi_1 + \Phi_2 + \Phi_3) y_{t-1} - (\Phi_2 + \Phi_3) \Delta y_{t-1} - \Phi_3 \Delta y_{t-2} + \varepsilon_t, \\
&\qquad \Rightarrow \; -y_{t-1} \text{ lhs / rhs} \\
\Delta y_t &= (\Phi_1 + \Phi_2 + \Phi_3 - I) y_{t-1} - (\Phi_2 + \Phi_3) \Delta y_{t-1} - \Phi_3 \Delta y_{t-2} + \varepsilon_t \\
&= \Pi y_{t-1} + \Gamma_1 \Delta y_{t-1} + \Gamma_2 \Delta y_{t-2} + \varepsilon_t,
\end{aligned}
$$

where $\Pi = \Phi_1 + \Phi_2 + \Phi_3 - I, \quad \Gamma_1 = -(\Phi_2 + \Phi_3), \quad \Gamma_2 = -\Phi_3,$

$$0 < \text{rank}(\Pi) < N.$$

General: VAR(p) to VECM

The same recursions can be used for a cointegrating VAR(p) model

$$y_t = \Phi_1 y_{t-1} + \ldots + \Phi_p y_{t-p} + \varepsilon_t,$$

to obtain

$$\Delta y_t = \Pi y_{t-1} + \Gamma_1 \Delta y_{t-1} + \ldots + \Gamma_{p-1} \Delta y_{t-p+1} + \varepsilon_t,$$

where

$$\Pi = -I + \sum_{j=1}^{p} \Phi_j, \quad \Gamma_i = -\sum_{j=i+1}^{p} \Phi_j, \quad \text{for } i = 1, \ldots, p-1,$$

with

$$0 < \text{rank}(\Pi) < N, \qquad \text{such that } \Pi = \alpha\,\beta' \text{ is valid.}$$

In case $\text{rank}(\Pi) = 0$, the VAR(p) model is non-stationary but no cointegration.

In case $\text{rank}(\Pi) = N$, y_t is stationary and cointegration has no meaning here.

Decomposition of Π

The decomposition of a rank deficient matrix Π, as in

$$\Pi = \alpha\, \beta',$$

where $N \times r$ matrix α has "speed of adjustment" vectors and $N \times r$ matrix β has "cointegrating" vectors, is not unique: α and β can take different values and still producing the same Π.

In example, cointegrating vector $\beta = (1, -\beta_c)'$ let $\beta' y_t = y_{1,t} - \beta_c y_{2,t}$ be stationary, but $\beta^* = w \cdot \beta$, scalar $w \in \mathbb{R}$, is also a cointegrating vector.

To side-step this issue, in most empirical research we impose a structure on β to obtain a "unique" cointegrating vector

$$\beta = \left[\begin{array}{c} I_r \\ \hline \beta \end{array} \right],$$

with $(N - r) \times r$ matrix $\overline{\beta}$ of unique values.

VAR representation of VECM

Finally, notice that the reverse also works !!
To illustrate, the VECM

$$\Delta y_t = \Pi y_{t-1} + \Gamma \Delta y_{t-1} + \varepsilon_t,$$

can be written as a VAR for y_t. However, it's a cointegrating VAR and it's not a VAR(1) but a VAR(2) !

$$\begin{aligned} y_t &= (I + \Pi) y_{t-1} + \Gamma \Delta y_{t-1} + \varepsilon_t \\ &= (I + \Pi + \Gamma) y_{t-1} - \Gamma y_{t-2} + \varepsilon_t \\ &= \Phi_1 y_{t-1} + \Phi_2 y_{t-2} + \varepsilon_t, \end{aligned}$$

where $\Phi_1 = I + \Pi + \Gamma$ and $\Phi_2 = -\Gamma$.

The VAR representation is typically used to compute forecast and impulse response functions for cointegrating systems.

Maximum likelihood estimation for VECM

Given that the VECM

$$\Delta y_t = \Pi y_{t-1} + \Gamma_1 \Delta y_{t-1} + \ldots + \Gamma_{p-1} \Delta y_{t-p+1} + \varepsilon_t,$$

is a stationary model, the MLE of the parameters in a VECM is carried out in the same way as a VAR(p) model. We refer to the set of slides on VAR models and notice that we can express the VECM as $\Delta y_t = B\, X_t + \varepsilon_t$ with

$$B = [\Pi, \Gamma_1, \ldots, \Gamma_{p-1}], \qquad X_t = \left(y'_{t-1}, \Delta y'_{t-1}, \ldots, \Delta y'_{t-p+1}\right)'.$$

We denote the MLEs as

$$\widehat{\Pi}, \quad \widehat{\Gamma}_1, \ldots, \widehat{\Gamma}_{p-1}, \quad \widehat{\Sigma}.$$

The maximized $\widehat{\ell}^{\max}$ is used generally for likehood-based testing and for AIC to determine p, similarly as for a VAR(p) model.

Determine rank of Π

Given the MLE $\widehat{\Pi}$, its (ordered) eigenvalues are given by

$$\widehat{\pi}_1 > \widehat{\pi}_2 > \ldots \widehat{\pi}_N,$$

In case matrix Π in the true VECM has $\text{rank}(\Pi) = r$, we should have all $\widehat{\pi}_{r+1}, \ldots, \widehat{\pi}_N$ having values close to zero, relative to $\widehat{\pi}_1, \ldots, \widehat{\pi}_r$.

To empirically test whether collectively $\pi_{r+1}, \ldots, \pi_N$ are zero, the econometrician Soren Johansen proposes two test-statistics:

- Trace test
- Maximum eigenvalue test

Trace test

The ***trace statistic*** is given by

$$\pi_{\text{trace}}(r) = -T \sum_{i=r+1}^{N} \log(1 - \widehat{\pi}_i).$$

This test-statistic has the null $H_0 : \text{rank}(\Pi) = r$ against the alternative $H_1 : \text{rank}(\Pi) > r$.

If the trace test statistic is sufficiently small, we accept the null.

Similar to unit root tests, $\pi_{\text{trace}}(r)$ has a nonstandard distribution (depending on T and deterministic terms). In software packages, critical values are obtained via Monte Carlo simulation.

Maximum eigenvalue test

The *maximum eigenvalue statistic* is given by

$$\pi_{\max}(r) = -T \log(1 - \widehat{\pi}_{r+1}).$$

This test-statistic has the null H_0 : $\operatorname{rank}(\Pi) = r$ against the alternative H_1 : $\operatorname{rank}(\Pi) = r + 1$. If it is sufficiently small, we accept the null.

If we have $r + 1$ cointegrating relations, $\widehat{\pi}_{r+1}$ is positive and $\log(1 - \widehat{\pi}_{r+1}) < 0$ so that $\pi_{\max}(r)$ is relatively large.
If we have r cointegrating relations, $\widehat{\pi}_{r+1}$ is close to zero so that $\pi_{\max}(r)$ is relatively small.

Its critical value is obtained via Monte Carlo simulation (depending on T and deterministic terms).

Determine rank of Π

Johansen has proposed

- Trace test
- Maximum eigenvalue test

and they can be used together. When they give contradictory conclusions, the maximum eigenvalue test statistic is usually given priority.

These tests are carried out sequentially, starting with $r = 0$, and repeated upto $r = N$.

Finally, residual diagnostic checking should be done in each step, make sure that residuals are

- only weakly (serially) correlated (so no omitted variables),
- not excessively heteroskedastic (affects size and power of tests),
- approximately Gaussian.

Johansen Estimation Method for VECM

Johansen has proposed a step-wise method for estimation:

1. Plot time series and carry out unit root testing: the variables can only be cointegrated when they ALL are integrated.
2. Select lag length of VECM via MLE for different p values, based on LR test or AIC statistics.
3. Estimate parameters for selected VECM and determine cointegrating rank r using $\pi_{\text{trace}}(r)$ and $\pi_{\text{max}}(r)$ tests.
4. Determine α and β matrices and make sure that they are conform to implications in economics, finance or other.
5. Residual diagnostic checking, forecasting, impulse response functions, etc.

Much more to do, for later

We have presented some key elements of cointegrating VAR models and VECMs.

We have concentrated on the methodology, we have not discussed theoretical aspects and we have not provided empirical illustrations.

More advanced econometrics courses will discuss the theoretical and technical aspects. An empirical illustration is covered in Assignment 3.

There are more issues that need attention. In particular, the incorporation of deterministic effects in the VECM, including intercept and trend functions.

What did we do ?

- VAR model and non-stationary processes
- We introduced Cointegration in the context of a VAR model
- Introducing Vector Error Correction Model (VECM)
- We reduced a test for Cointegration to a rank test in a VECM
- Introducing the Johansen methodology